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Business Understanding Report

Commercial Objectives, Problem Formulation & Retail Strategy for ShopSmart Ltd.

Client / Partner:	ShopSmart Retail Ltd.
Curriculum Track:	Week 2: Data Preprocessing & Feature Engineering
Author:	Junior Machine Learning Engineer
Dataset Scope:	50,000 Transactions, 25 Stores, Nationwide Nigerian Footprint
Primary Targets:	Order Basket Total (■) & Product Return Probability (0/1)

EXECUTIVE MANDATE:
ShopSmart Retail Ltd. operates in an increasingly competitive retail landscape across 25 metropolitan stores and an expanding eCommerce channel. While top-line sales exceeded ■7.7 Billion across 50,000 historical orders, the business faces dual margin headwinds: unpredictable basket sizes and reverse-logistics friction from 2,442 returned orders (4.88% return rate). This report translates high-level retail goals into concrete machine learning problem formulations, aligning engineering pipelines with financial ROI.

1. Business Background & Market Context

ShopSmart Retail Ltd. is an established Nigerian omnichannel retailer providing products across 5 primary departments: Groceries, Fashion, Electronics, Home, and Beauty. As retail volume migrates to digital channels, transaction dynamics have grown complex. Pricing markups, shipping subsidies, customer loyalty tiers (Bronze, Silver, Gold), and regional supply chains directly dictate whether a transaction yields sustainable margin or operational loss.

2. Strategic Business Objectives & Machine Learning Mapping

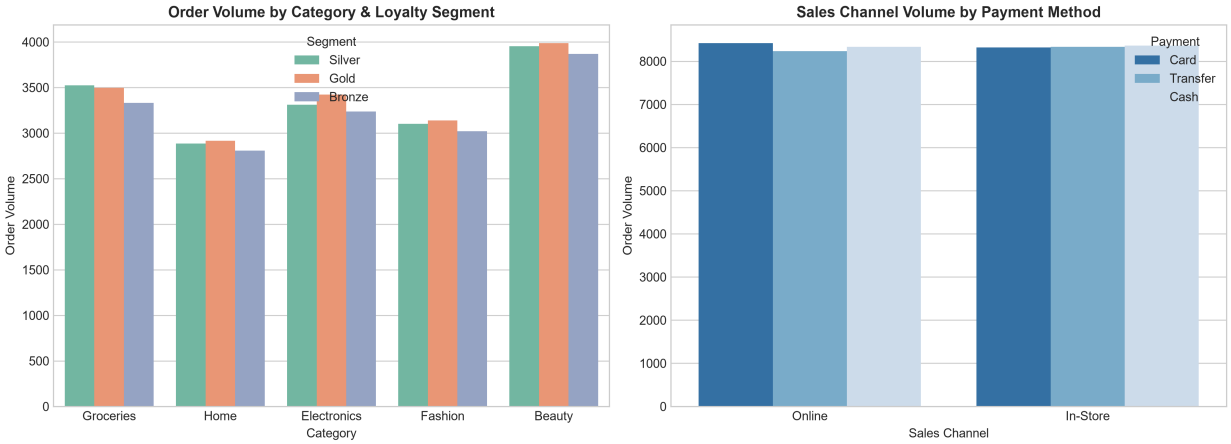
Strategic Business Goal	Operational Pain Point	ML Task & Target Variable	Target ROI Metric
Revenue Maximization & Upselling	Uncertain basket sizes; static discounts eroding margin without lifting total cart volume.	**Regression:** Predict continuous Order_Total (■) at point of checkout.	+8-12% uplift in Average Order Value (AOV); optimized bundle promotions.
Reverse-Logistics Loss Mitigation	2,442 returns creating restocking costs, repackaging delays, and customer friction.	**Classification:** Predict binary Return_Flag (0 = Kept, 1 = Returned).	15-20% reduction in reverse transit waste via proactive dispatch packaging.

3. Commercial Stakeholder Matrix & Value Realization

To ensure operational adoption, preprocessing and feature engineering must serve specific departmental stakeholders:

- **Chief Commercial Officer (CCO):** Needs accurate basket spend forecasts to calibrate loyalty tier thresholds and promotional margin ceilings.
- **Head of Supply Chain & Logistics:** Demands automated return risk scoring to route high-risk orders (e.g. fragile Electronics and Home items) through reinforced packaging and courier tracking.
- **Store Regional Managers (Lagos, Ibadan, Abuja, etc.):** Require localized demand visibility to manage regional inventory holding levels and restock schedules.
- **Customer Experience (CX) Leads:** Require early notification when high-value orders encounter delivery friction, protecting customer lifetime value (CLV).

4. Key Observations from Initial Commercial Exploration



Commercial Takeaways: Order volumes across categories show balanced purchasing, but customer loyalty segments behave distinctly. Gold tier shoppers contribute over ■2.61 Billion in cumulative sales. Payment methods are heavily dominated by Card and Bank Transfer across both In-Store and Online channels. Online transactions experience higher shipping friction, necessitating explicit shipping ratio feature engineering.

5. Preprocessing Strategy & Pipeline Requirements

Raw transactional logs cannot be fed directly into estimators. To guarantee predictive performance without data leakage or bias, our engineering roadmap mandates: **(1) Strict Data Hygiene:** Reconciling order totals and removing PII keys (names, emails, phones); **(2) Outlier Winsorization:** Capping extreme financial spikes to preserve all 50,000 records without distortion; **(3) Multi-Scale Transformation:** Applying RobustScaler to skewed features and StandardScaler to normal features; and **(4) Dimensionality Selection:** Filtering low-variance noise while preserving high-impact profit and interaction features.